



Data Driven Education

Discussion Forum Subteam

Presentation 1 - 09/24/2025

Meet the Team



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Introduction

- Online Discussion platform is crucial in modern education and digital studies.
- **Ed Discussion and EdX:** widely used online discussion forum
 - EdX offers Massive Open Online Courses (MOOCs)
 - enables students and instructors to interact
 - discuss course materials asynchronously
- What are some unique characteristics of Online Discussion Forums?
- What to do to utilize Online Discussion platforms better?
- Improve the whole education experience

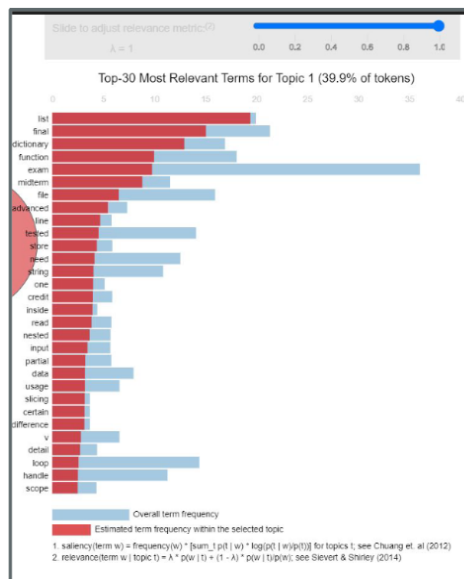
Research Questions

- How to combine sentiment analysis and cognitive presences to provide insights into student engagement and attitudes toward the course?
- How to use machine learning techniques to predict students' success in the course based on their discussion engagements?
- What is the best way to present discussion analytics to instructors in an actionable and user-friendly dashboard?

Spring 2025 Recap:

Data Science:

- Classification
 - Focus on text > metadata classification
 - EdX DF dataset
- Analysis and Interpretation
 - Latent-Dirichlet Allocation (LDA) to measure token-frequency
 - Gensim, PyLDAvis, and Matplotlib



Machine Learning:

- Outcome
 - Developed a synthetic CS 1301 dataset using OpenAI's o1 model
 - Topic distribution over time

Mock-up React dashboard

Discussion Forum - Ed Discussion Topic Analysis

Topic Selection

CS 1301 - topic 1

CS 1301 - topic 2

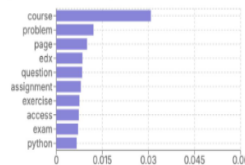
CS 1301 - topic 3

CS 1301 - topic 4

CS 1301 - topic 5

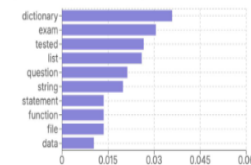
Analysis with Real Dataset

Top Words for CS 1301 - topic 5:



Analysis with Synthetic Dataset

Top Words for Synthetic data - topic 5:



Explanation of Dashboard

This is the dashboard for visualizing the output for discussion forum with LDA topic modeling.

Left side shows the result from real dataset while the right side shows the result from synthetic dataset we generated from our machine learning model.

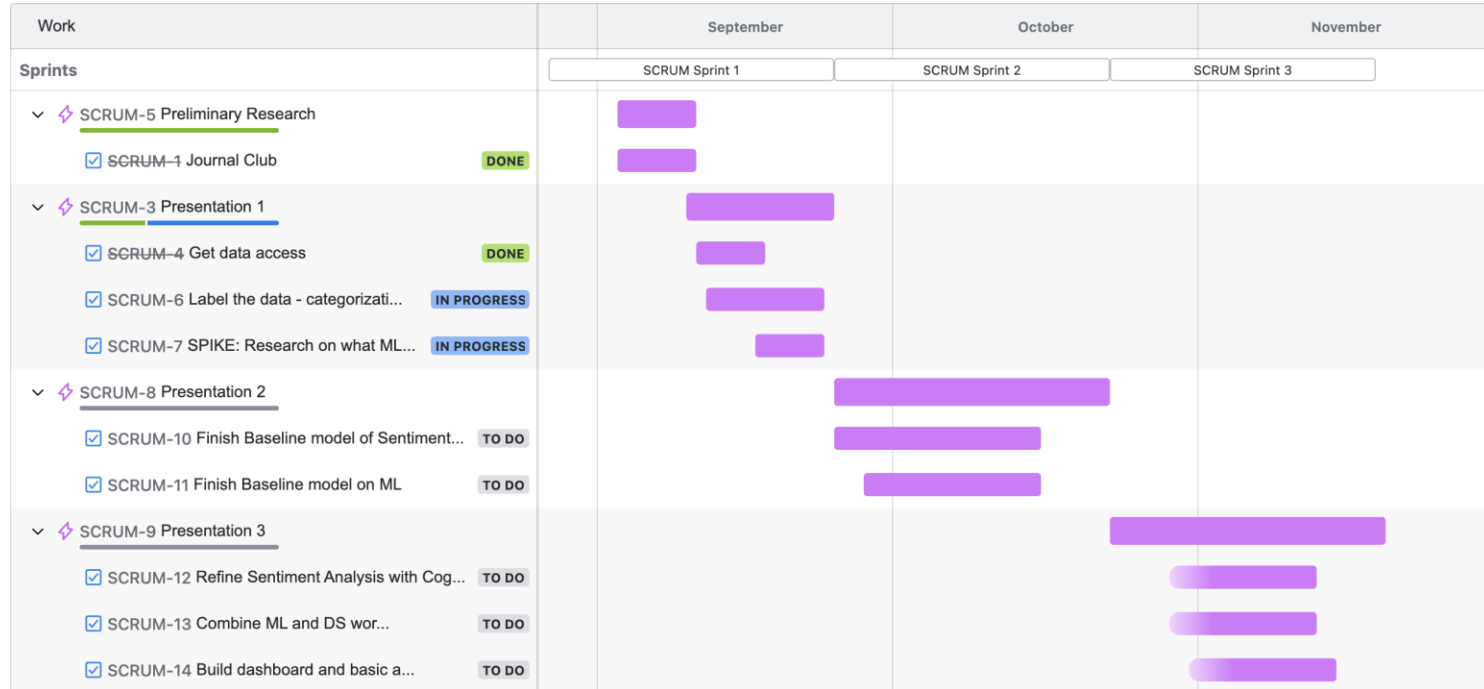
Topic	Subtopics	Weeks Covered	Learning Outcomes	Student Feedback
Variables and Data	['Integers', 'Strings', 'Booleans']		1 Understand basic data types and variables	I'm unsure when to convert floats to ints for accurat
Loops and Condition	['For Loops', 'If Statements']		2 Use loops for repetition and conditions	For the midterm, do I need to memorize nested loop
Functions	['Defining Functions', 'Parameters']		2 Define and use functions in Python	Can we have a function return more than one list wi
Data Structures	['Lists', 'Dictionaries']		2 Use basic data structures	Is it better to store large sets of data in a dictionary
File I/O	['Reading Files', 'Writing Files']		1 Read from and write to files in Python	Do we need to handle exceptions for every file opera
Variables and Data	['Integers', 'Strings', 'Booleans']		1 Understand basic data types and variables	Is there any real performance difference between in
Loops and Condition	['For Loops', 'If Statements']		2 Use loops for repetition and conditions	How do I break out of multiple nested loops without
Functions	['Defining Functions', 'Parameters']		2 Define and use functions in Python	Would using *args for variable-length parameters c
Data Structures	['Lists', 'Dictionaries']		2 Use basic data structures	Is it possible that we get tested on dictionary compr
File I/O	['Reading Files', 'Writing Files']		1 Read from and write to files in Python	Is 'with open' mandatory, or can I just open and clo
Variables and Data	['Integers', 'Strings', 'Booleans']		1 Understand basic data types and variables	How do I handle a boolean variable for a flag in an
Loops and Condition	['For Loops', 'If Statements']		2 Use loops for repetition and conditions	Does indentation have a big effect on partial credit
Functions	['Defining Functions', 'Parameters']		2 Define and use functions in Python	Are lambda functions allowed on the test or should
Data Structures	['Lists', 'Dictionaries']		2 Use basic data structures	I keep forgetting the difference between list.pop() ar
File I/O	['Reading Files', 'Writing Files']		1 Read from and write to files in Python	If we read a file line by line, will that appear on the

Goals for the Semester

- Enhance analysis of online discussion forum data
 - Data-driven insights, ML models, and more
 - Develop a dashboard
- Explore sentiment analysis with cognitive presences in addition to topic modeling
 - Utilizing BERT
- Build a model to analyze students' grades relations to online engagement



Project Timeline



Jira - For task tracking and management

Data Science

Goals & Timeline for the Semester

Spring 2025

- January – February:
 - Learn **data cleaning & processing** techniques
- February – April:
 - Create text-analysis models for **sentiment analysis** via LDA & TF-IDF
- April – May:
 - Revamp dashboard from last semesters



Fall 2025

- August – mid-October:
 - Continue exploring text-analysis models for **sentiment analysis**
 - **BERT** - Google's NLP model that can learn a wide breadth of tasks relating to language
- Mid-October – December:
 - Analyze dataset to understand student levels of **cognitive presence**
- December:
 - Using insights, revamp dashboard from last semesters

Metrics/Data We're Using/ Why this Data is Relevant

Hybrid approach using preprocessed and processed edX data

- ~18.5k data entries
- Tradeoff assessment
- Determine from results the more useful data

BERT - Deep learning model

Sprint 1: Benchmark

- Label the first thousand data points based on five declared, sentiment categories

Sprint 2: Scaling

- Create **data pipeline** optimized for BERT to produce sentiment analysis results at scale
- Fine-tune model for **predictive** analysis

Sprint 3: Interpret

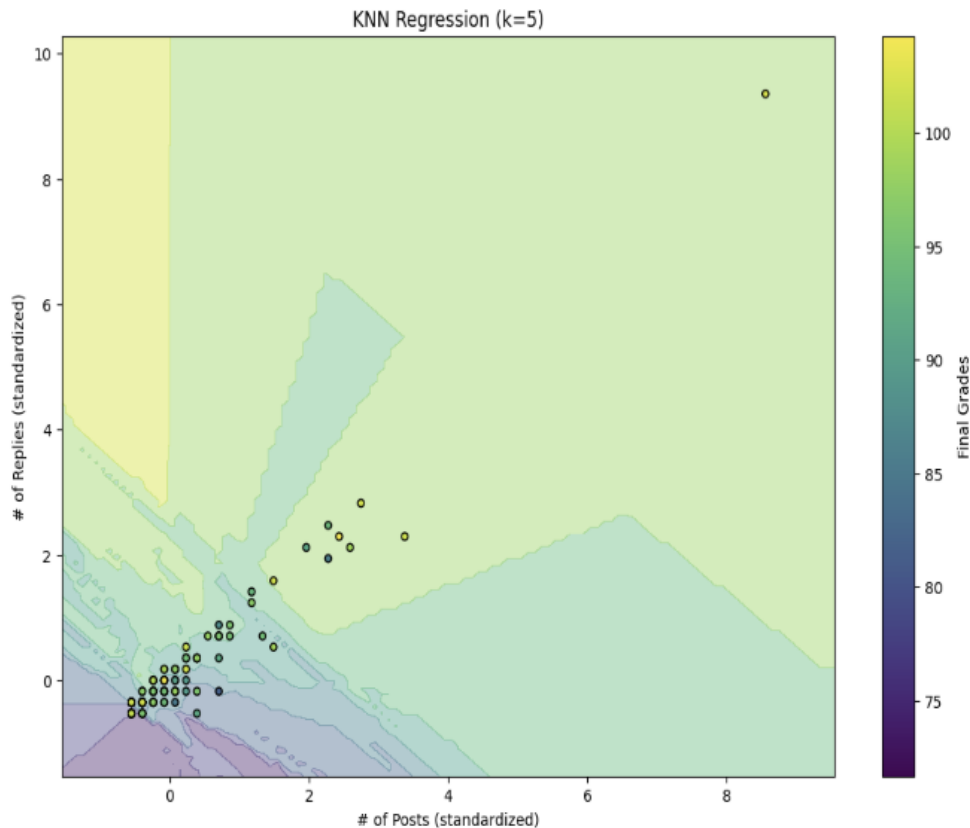
- Integrate data viz tools into user-friendly **dashboard** + deployment?

Machine Learning

Spring Recap:

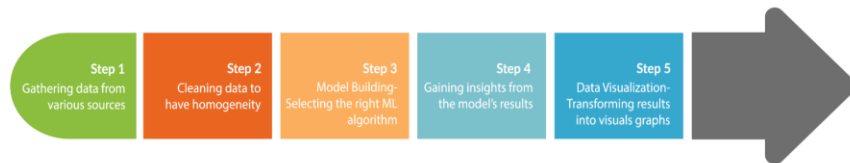
KNN Model:

- K-Nearest Neighbor Model uses k-nearest points to find to predict the response variable with n factors
- We tuned this model and found the lowest error when using $k=5$
- Replying to posts reflects more positively on grades than creating new posts



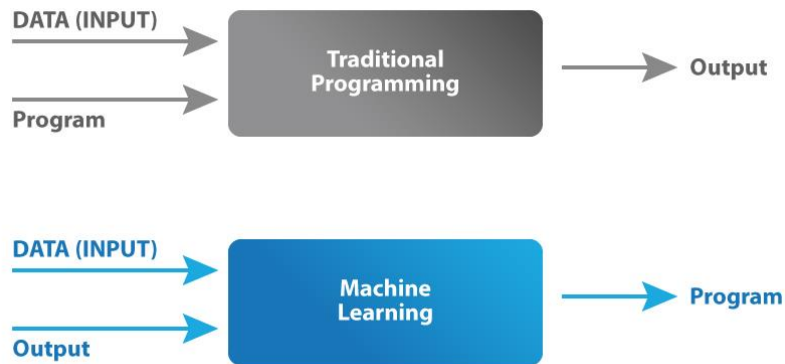
Goals for the Semester

- Using data from CS1301 2017 & 2018
 - Join grades, forum post, and forum thread datasets with queries
 - Clean null/blank information
- Process data to create a bi-modal student behaviors model
 - Our target variable is student grades and we want to identify which forum behaviors predict academic success
 - We want to study 2 topics:
 - Text embeddings (semantics, etc.)
 - Time-series (frequency, timing in sem, etc.)
- Once identifying which behaviors are key after fusing our bi-modal model, we can create visualizations of those traits and their correlation to grades



Progress So Far:

- Identified usable data for this model
- Concluded the two best modalities to consider when assessing student behaviors with our dataset
- Decided to use LSTM ML function for time-series modality and Transformer ML function for semantic modality
- Started writing queries to join 3 tables where necessary for end model to process



Literature Review

Literature Review:

Goals:

- Understand how sentiment analysis and cognitive presence overlap to provide insights on student engagement and attitudes towards the course
- Explore prior applications of learning analytics dashboards to identify best practices for integrating data visualization into actionable tools for instructors.
- Examine engagement metrics and labeling strategies from past studies to guide how we categorize sentiment and measure student participation in our project.

Body of Knowledge:

- “Analyzing Learning Sentiments on a MOOC Discussion Forum Through Epistemic Network Analysis.” (Jianhui Yu, 2025)
 - Examines how sentiments evolve over time in MOOCs, and their relationships with performance and topic.
- “From Concept to Classroom: Developing Instructor Dashboards through Human-Centered Design.” (Alzoubi, Dana, et al., 2024)
 - Describes a human-centered design approach for instructor dashboards.

Literature Takeaways:

"Analyzing Learning Sentiments on a MOOC Discussion Forum Through Epistemic Network Analysis." (Jianhui Yu, 2025)

- Epistemic network analysis (ENA) can map co-occurrence of different sentiment states
- Identify variation in sentiment patterns by performance group
- Take into account sentiment frequency and changes in those over time.

"From Concept to Classroom: Developing Instructor Dashboards through Human-Centered Design." (Alzoubi, Dana, et al., 2024)

- Human-centered design should be a priority
- Focus on providing actionable insights over raw data
- Refine the dashboard with instructor input over multiple iterations
- Ensure usability: complex results should be translated into simple engagement metrics

Conclusion

- Improve course design and refine teaching strategies
- Provide inclusive learning environments
- Personalize learning for foreseeable weakness
- Promote student's engagements and grades
- Ensure a more responsive and adaptive online learning experience.

Citations

1. Alzoubi, Dana, et al. "From Concept to Classroom: Developing Instructor Dashboards through Human-Centered Design." *Computers and Education Open*, vol. 7, Dec. 2024, Article 100234, doi:10.1016/j.caeo.2024.100234.
2. Yu, Jianhui. "Analyzing Learning Sentiments on a MOOC Discussion Forum Through Epistemic Network Analysis." *The International Review of Research in Open and Distributed Learning*, vol. 26, no. 1, Mar. 2025, pp. 197-215, doi:10.19173/irrodl.v26i1.7965.

Thanks for listening!

Q&A